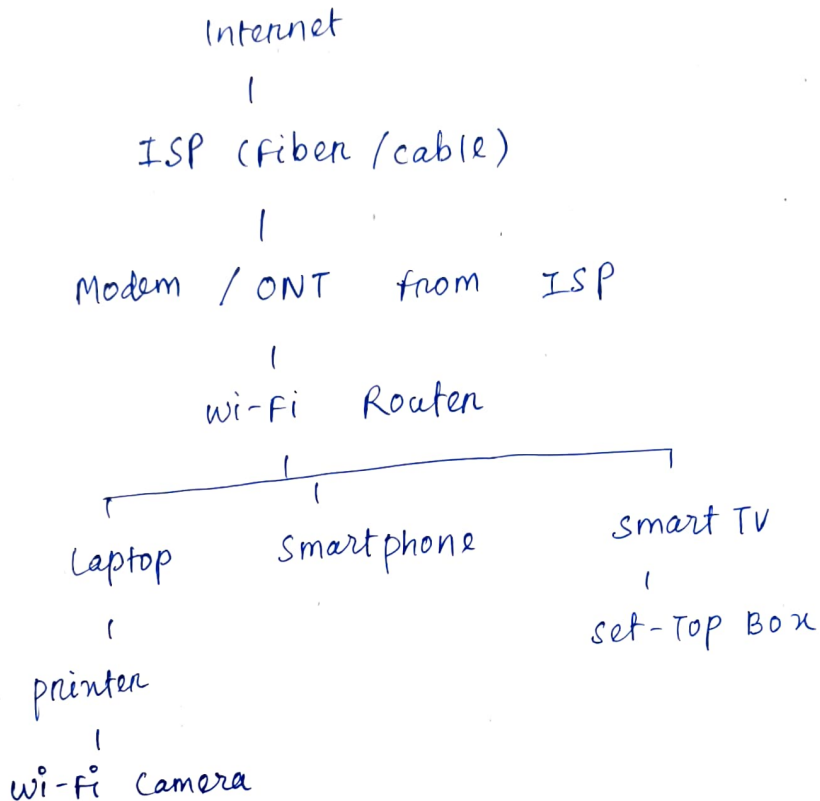


CN Assignment - 1: Draw your Home Network Topology and explain it.

Network Topology used: Star Topology



(i) Internet source (ISP - Internet service provider):

The internet service is provided by an ISP such as Airtel, Jio, BSNL etc. The ISP delivers internet through a fiber optic or coaxial cable that connects to a modem or an ONT (Optical network Terminal) at our home.

(ii) Modem / ONT:

The modem or ONT (depending on whether it's cable or fiber) is the first device that receives the internet signal. It converts the incoming

line signal into digital data that can be transmitted to the router.

3) Wi-Fi Router

This is the central hub of the home network. It connects to the modem and creates a local Area Network (LAN) by distributing the internet connection wirelessly (Wi-Fi) or through Ethernet cables.

Router Functions:

- Assigns IP addresses to devices in the network (using DHCP)
- provides network security through firewalls and password protection.
- can manage bandwidth and parental controls

4) Connected Devices:

- Laptop/Desktop computer : connected via Ethernet cable or Wi-Fi for stable internet access, work, browsing, etc.
- Smartphone/Tablet : connects via Wi-Fi to access internet services like browsing, social media and apps.
- Smart TV : streams content from platforms like YouTube, Netflix via Wi-Fi
- Set-Top Box (OTT or IPTV) : uses internet connection for on-demand video streaming.

- wi-fi printer: can print wirelessly from any device connected to the network.

- wi-fi security camera: sends live footage over the internet and can be accessed remotely.

Topology Type: star Topology

- All devices are independently connected to the central router.
- If one device fails, it does not affect others.
- Easy to manage, expand and troubleshoot.

Benefits of Home Network

- centralized management
- Easy sharing of printing and files between devices.
- Support multiple smart home devices.
- Enables internet access for work, study and entertainment.